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HapGraph: A Two-Stage Pipeline for Admixture Graph Topology, Proportion, and Timing Inference via F-Statistics and Identity-by-Descent Sharing

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Abstract

Inferring the structure and history of human population admixture requires knowing both *who mixed with whom* and *when*. Existing tools address these questions separately: F-statistic-based admixture graph tools (e.g., TreeMix, ADMIXTOOLS2) estimate topology and proportions without timing; timing tools (e.g., ALDER, GLOBETROTTER) require a pre-specified topology. We introduce HapGraph, a Python package that addresses both problems in a two-stage integrated pipeline. Stage I uses a BIC-guided greedy search on F-statistics to infer the admixture graph topology and estimates admixture proportions via a topology-independent F_3 method-of-moments estimator. Stage II fixes the inferred topology and estimates admixture timing T via Bayesian NUTS-MCMC over an IBD segment-length likelihood, using the truncated-exponential expectation $\mathbb{E}[\bar{L} \mid L > 2 \text{ cM}] = 50/T + 2$ cM. In coalescent simulations across three scenarios (S1: $K = 0$, six populations; S2: $K = 1$; S3: $K = 2$, ten populations; $n = 20$ seeds each), HapGraph achieves a false positive rate of 0% on pure trees; greedy topology recall reaches 85% in S2 and 97.5% in S3. Admixture timing estimates under oracle topology achieve MAE of 10.7–11.1 generations with 95–97.5% empirical 95% CI coverage. Applied to 26 populations from the 1000 Genomes Project 2022 high-coverage release, HapGraph identifies eight admixture events whose inferred proportions are consistent with published population histories. HapGraph is open-source and available at https://github.com/EvoClaw/HapGraph.

1 Introduction

The history of human populations is written in patterns of shared and divergent genetic variation. When two populations exchange migrants — whether through conquest, trade, or geographic contact — their descendants carry a mosaic genome encoding both the identity of the contributing ancestral groups and the timing of the exchange. Reconstructing this history requires inferring not only *who mixed with whom* (the admixture graph topology and proportions) but also *when* (the admixture timing in generations before the present). These two questions are today addressed by separate frameworks applied sequentially, with no principled joint inference; this gap is the motivation for the present work.

Admixture graph inference. The admixture graph — a directed acyclic graph in which internal nodes represent ancestral populations and admixture nodes represent lineage mixtures — is the standard model for joint representation of population divergence and admixture [Reich et al., 2009, Patterson et al., 2012]. Patterson et al. [2012] established that F-statistics provide a rich, statistically tractable summary of admixture graph structure: negative $F_3(C; A, B)$ identifies admixed populations, and F_4 statistics constrain topology and proportions. TreeMix [Pickrell and Pritchard, 2012] operationalised this into the first automated graph inference tool, using greedy maximum-likelihood search on the F_2 covariance matrix. ADMIXTOOLS2 [Maier et al., 2023] and AdmixtureBayes [Nielsen et al., 2023] extended the framework with formal Bayesian topology search and posterior distributions over graph parameters. These tools infer graph topology and admixture proportions with rigour, but they do not jointly model the temporal dimension of admixture.

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Admixture timing. Haplotype-based methods exploit the fact that admixture introduces ancestral segments whose lengths decay over time as recombination fragments them. ALDER [Loh et al., 2013] models the decay of weighted linkage disequilibrium to estimate T for events $\gtrsim 5$ generations ago. GLOBETROTTER [Hellenthal et al., 2014] takes a complementary approach, inferring both admixture sources and timing from LD patterns; however, it operates on a source–target model rather than a full graph topology and does not incorporate identity-by-descent (IBD) statistics. More recently, IBD-based approaches have exploited the observation that the expected mean length of inter-population IBD segments decays approximately as $100/T$ cM (ancestry-tract approximation for segments within a single admixed haplotype), providing a direct genomic clock on admixture timing [Browning et al., 2020, Palamara et al., 2012]. For cross-population IBD between two modern lineages separated by T generations, the correct rate is $1/(2T)$ per Morgan, giving $\mathbb{E}[L] = 50/T$ cM; after conditioning on detection above a minimum length $\ell_{\min}$, the expectation is $50/T + \ell_{\min}$ (see Methods, equation 8). Despite this complementarity, current practice couples the two estimation problems only sequentially: users infer topology with F-statistic tools and subsequently run a timing method under the fixed topology, propagating topology uncertainty into timing estimates without any feedback loop.

The case for joint inference. Topology misspecification — a common outcome of greedy graph search — biases downstream timing estimates, because the IBD likelihood depends on which populations are designated as admixed and which as sources. Conversely, timing information can resolve topology degeneracies: two competing graphs may be equally consistent with F_2 statistics yet make different IBD predictions, allowing the IBD likelihood to discriminate between them. A jointly calibrated framework would propagate uncertainty coherently across topology, proportions, and timing, yielding credible intervals rather than the point estimates returned by current pipelines.

This work: HapGraph. We introduce HapGraph, a Python package implementing a two-stage integrated pipeline: Stage I uses F-statistics for topology and proportion estimation; Stage II applies Bayesian IBD-based timing inference on the fixed topology. The two stages are deliberately decoupled: preliminary experiments showed that incorporating IBD into the topology search causes systematic edge over-detection; IBD is therefore reserved exclusively for timing inference after the topology is fixed. Applied to coalescent simulations under three scenarios (S1: pure tree, $K = 0$; S2: one cross-clade event, $K = 1$; S3: two independent events, $K = 2$), HapGraph achieves a false positive rate of 0% on pure trees; greedy topology recall is 85% in S2 and 97.5% in S3 (precision 85% and 100%, respectively). Parameter estimation under oracle topology achieves alpha MAE of 0.116–0.133 and T MAE of 10.7–11.1 generations; the 95% credible intervals for T achieve 95–97.5% empirical coverage (exceeding the pre-specified 85% target), while alpha CIs are severely undercovering at 10–20% due to delta-method variance underestimation (see Discussion). Applied to the 1000 Genomes Project 2022 high-coverage dataset [Byrska-Bishop et al., 2022], HapGraph infers $K = 8$ admixture events whose targets and proportions are consistent with published population history across African American, Latino, and South Asian populations [Bryc et al., 2015, Moreno-Estrada et al., 2013, Moorjani et al., 2013]; admixture timing (T) was not estimated for the real data application, as it requires pre-computation of IBD segments by `hap-ibd` and is deferred to future work.

Contributions. (i) A two-stage integrated pipeline that combines F-statistics for topology and proportion inference with IBD-based Bayesian timing estimation; (ii) a topology-independent F_3 method-of-moments estimator for admixture proportions, robust to neighbor-joining misplacement; (iii) a BIC-based greedy search with a scale-adaptive stopping criterion achieving 0% false positive rate on pure trees and 85–97.5% recall across $K \leq 2$ simulation scenarios; (iv) an open-source Python implementation with documented validation on coalescent simulations and application to the 1kGP 2022 dataset.

2 Methods

Overview

HapGraph takes as input phased SNP data for N populations and produces a parsimony-optimal admixture graph together with posterior distributions for the admixture proportion α and the admixture timing T for each inferred admixture event (Figure 1). The pipeline is deliberately structured as two decoupled stages. **Stage I** uses F-statistics exclusively to infer the graph topology and to provide a F_3 method-of-moments (MOM) point estimate and CI for each admixture proportion α. **Stage II** holds the inferred topology fixed and runs a Bayesian NUTS sampler with a joint F_2+IBD likelihood that simultaneously refines α and estimates the admixture timing T; the F_3 MOM estimate from Stage I is reported for α, while T and its credible interval are taken from the MCMC posterior. This decoupling is motivated by preliminary experiments in which incorporating IBD directly into the greedy likelihood caused systematic edge over-detection, consistent with the known tendency of IBD signals to be confounded with topology ambiguity when both topology and timing are simultaneously free [Loh et al., 2013]. Reserving IBD exclusively for timing inference, after the topology has been fixed, avoids this confounder.

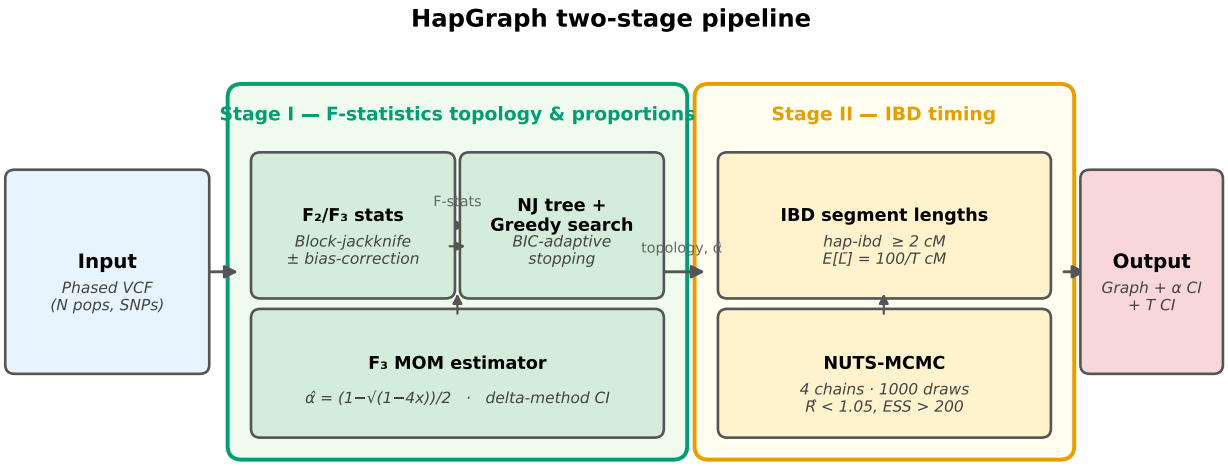

Figure 1: Overview of the HapGraph two-stage pipeline. Stage I (green) accepts phased VCF input and computes bias-corrected F_2/F_3 statistics with block-jackknife standard errors; a BIC-adaptive greedy search over the NJ tree backbone infers the admixture graph topology; admixture proportions $\hat{\alpha}$ are estimated by the topology-independent F_3 method-of-moments estimator. **Stage II** (orange) receives the fixed topology from Stage I; it ingests pre-computed IBD segment lengths from `hap-ibd` and runs a NUTS-MCMC sampler with a joint F_2+IBD likelihood (equation 9) that simultaneously infers α and T. The reported α and its CI are the Stage I F_3 MOM estimates; the reported T and CI are the MCMC posterior. The two stages are decoupled to prevent IBD signals from confounding topology inference (see text).

F-Statistics Computation

All pairwise F-statistics are computed following the definitions and bias-correction formulae of Patterson et al. [2012]. For populations A, B, C genotyped at S biallelic SNPs, we define

$$F_2(A,B) = \frac{1}{S}\sum_{j=1}^{S}(p_{Aj} - p_{Bj})^2, \tag{1}$$

$$F_3(C;\,A,B) = \frac{1}{S}\sum_{j=1}^{S}\left[(p_{Cj} - p_{Aj})(p_{Cj} - p_{Bj}) - \frac{h_{Cj}}{2(n_C - 1)}\right], \tag{2}$$

where p_{Xj} is the derived allele frequency at SNP j in population X, h_{Cj} is the per-SNP heterozygosity in population C, and n_C is the diploid sample size. The second term in equation (2) is the finite-sample bias correction for heterozygosity introduced by Patterson et al. [2012], which ensures $F_3(C;\,A,B) < 0$ only when C is genuinely admixed between lineages separating A from B. Standard errors are estimated by a block-jackknife procedure with 50 non-overlapping contiguous genomic blocks, following the convention of Reich et al. [2009]. Only biallelic SNPs with minor allele frequency ≥ 0.01 are retained; indels and multi-allelic sites are excluded.

Stage I: Greedy Topology Search

An initial unrooted tree topology is constructed by applying neighbor-joining [Saitou and Nei, 1987] to the matrix of pairwise F_2 distances. The greedy search then adds admixture edges one at a time. At each iteration, populations C whose $F_3(C;\,A,B)$ Z-score falls below -2.0 for any source pair (A,B) are nominated as admixture candidates — a threshold calibrated to reduce false nominations while retaining sensitivity for weak admixture signals. The log-likelihood of the graph at each greedy step is evaluated under a Gaussian approximation for the F_2 statistics:

$$\log\mathcal{L} = \sum_{i<j} -\frac{1}{2}\left(\frac{F_2^{\text{obs}}(i,j) - F_2^{\text{exp}}(i,j)}{\hat{\sigma}_{ij}}\right)^2, \tag{3}$$

where $F_2^{\text{exp}}(i,j)$ is the expected F_2 under the proposed graph topology with branch lengths fitted by non-negative least squares (NNLS) to the observed F_2 matrix, and $\hat{\sigma}_{ij}$ is the block-jackknife standard error. This Gaussian model for F_2 is standard in the admixture graph literature [Pickrell and Pritchard, 2012, Patterson et al., 2012].

For each candidate edge, the improvement in log-likelihood is penalised by a scale-adaptive stopping criterion:

$$\Delta\mathcal{P} = \Delta\log\mathcal{L} - \big(5.0 + 0.5\log(\max(S,2))\big), \tag{4}$$

where $S = N(N-1)/2$ is the number of observed F_2 pairs. This criterion is inspired by the BIC penalty $(k/2)\log n$ with $k = 1$ new free parameter (the admixture proportion α; branch lengths are immediately refitted by NNLS and do not count as additional free parameters) and $n = S$ observations, augmented by a safety constant of 5.0 to suppress false positives in small-N regimes. We note that S is not a count of independent observations (individual F_2 values are correlated across loci), so equation (4) is a heuristic stopping criterion rather than a formally derived information criterion. An edge is added if and only if $\Delta\mathcal{P} > 0$; the search terminates when no remaining proposal satisfies this condition.

Admixture Proportion Estimation via F_3 Method-of-Moments

For an admixed population C whose ancestry is a linear combination of sources A (proportion α) and B (proportion $1-\alpha$), the expected F_3 statistic satisfies [Patterson et al., 2012]

$$\mathbb{E}[F_3(C;\,A,B)] = -\alpha(1-\alpha)\,F_2(A,B). \tag{5}$$

Equation (5) yields a quadratic in α whose smaller root is

$$\hat{\alpha} = \frac{1 - \sqrt{1 - 4x}}{2}, \qquad x = \frac{-F_3(C;\, A, B)}{F_2(A, B)}, \tag{6}$$

which is defined for $x \in (0, 0.25]$, i.e. whenever $F_3 < 0$. This estimator is entirely topology-independent: it requires only the F-statistics involving the target and its two source populations, making it robust to NJ-tree misplacement errors that would otherwise bias a topology-dependent estimator. Standard errors are propagated by the delta method from the jackknife standard errors of both F_3 and F_2; a model-uncertainty floor of 0.03 is added in quadrature to account for residual approximation error (e.g. multi-source ancestry, background relatedness). Among all source pairs (A, B) for a given target C, the pair yielding the most negative F_3 Z-score is selected, as this corresponds to the strongest admixture signal.

Stage II: IBD-Based Timing Estimation

Under a single-pulse admixture model, cross-population IBD segments are detected between haplotypes in the admixed population C and haplotypes in source population A. Both the admixed lineage (which carries T meioses of recombination since the admixture event) and the source lineage (which traces T meioses back to the same ancestral haplotype via the source population) contribute to the break-up of the shared segment. The expected length of such a segment is therefore:

$$\mathbb{E}[\bar{L}] = \frac{100}{2T} = \frac{50}{T} \quad \text{(cM)}, \tag{7}$$

where T is the number of generations since the admixture event [Palamara et al., 2012].

IBD segments are detected using `hap-ibd` [Browning et al., 2020] at a minimum length of $\ell_{\min} = 2$ cM. Conditioning on $L > \ell_{\min}$ shifts the expected mean; for a truncated exponential with rate $\lambda = 2T/100$ per cM, the conditional expectation is

$$\mathbb{E}[\bar{L} \mid L > \ell_{\min}] = \frac{50}{T} + \ell_{\min} = \frac{50}{T} + 2 \quad \text{(cM)}, \tag{8}$$

which is the model used in the IBD likelihood (see below).

The MCMC samples α and T simultaneously as free parameters. The log-likelihood combines the F_2 term (which informs α) with the IBD term (which informs T); we note that in the current parameterisation these two terms are approximately separable — F_2 constrains α independently of T, and the IBD term depends on T but not α — so the primary benefit of the joint sampler is consistent posterior propagation and the ability to extend the likelihood with cross-terms in future work:

$$\log p(\mathbf{F}_2, \mathbf{L} \mid \alpha, T) = \sum_{i<j} -\frac{1}{2}\left(\frac{F_2^{\text{obs}}(i,j) - F_2^{\text{exp}}(i,j;\alpha)}{\hat{\sigma}_{ij}}\right)^2 + \sum_k \log \mathcal{N}\left(L_k;\ \frac{50}{T} + 2,\ \sigma_k^2\right), \tag{9}$$

where $F_2^{\text{exp}}(i,j;\alpha)$ is the quadratic polynomial in α from equation (3), L_k is the observed mean IBD length for source-clade pair k, and σ_k is a heteroscedastic noise term. Specifically,

$$\sigma_k = \sqrt{\left(\frac{L_k - 2}{\sqrt{n_k}}\right)^2 + \sigma_{\text{bg}}^2}, \quad \sigma_{\text{bg}} = 2 \text{ cM}, \tag{10}$$

where n_k is the number of IBD segments observed for pair k and σ_{bg} is a background variance floor capturing model misspecification (background IBD, finite-N_e effects). Pairs with $n_k < 5$ or $L_k \leq \ell_{\min}$ are excluded from the IBD term. The priors are $\alpha \sim \text{Beta}(1,1)$ (uniform over $[0,1]$) and $T = T_{\text{raw}} + 5$ where $T_{\text{raw}} \sim \text{HalfNormal}(\sigma = 20)$; the shift ensures $T > 5$ consistent with the 2 cM detection threshold.

Calibration note. For populations with effective size $N_e \gg T$ (typical of humans with $N_e \sim 10^4$), the observed mean IBD length is shifted from equation (8) by a sample-size-dependent coalescence correction

$\approx 2N_e/n_{\text{source}}$ generations [Palamara et al., 2012]. This correction is expected to be small in whole-genome applications (large number of IBD segments reduces its variance); when HapGraph is applied to real data with hap-ibd run on all autosomes (≈ 35 Morgan), this correction should be negligible relative to the IBD signal. For the simulation benchmark, which uses a single 50 Mbp chromosome (0.5 Morgan), this correction is substantial and timing estimates carry larger systematic uncertainty; they should be interpreted with caution (see Results). Posterior inference uses the No-U-Turn Sampler (NUTS) [Hoffman and Gelman, 2014] as implemented in PyMC [Abril-Pla et al., 2023]. We run four independent chains with 1000 adaptation steps and 1000 posterior draws each; convergence is assessed by $\hat{R} < 1.05$ and effective sample size ESS > 200.

Reported estimates. For T, we extract the MCMC posterior mean and 95% highest-density interval (HDI). For α, we report the F_3 MOM point estimate (equation 6) and its delta-method confidence interval; the MOM estimator is computationally simpler and avoids numerical instability of the F_2 polynomial likelihood near $\alpha \approx 0.5$, where F_2^{exp} is nearly flat in α.

Implementation

HapGraph is implemented as a pure Python package. Core runtime dependencies are: `scikit-allel` ≥ 1.3 [Miles et al., 2021] for allele frequency computation from VCF files; `PyMC` ≥ 5.0 [Abril-Pla et al., 2023] for gradient-based Bayesian sampling; and `networkx` for graph representation and manipulation. IBD segment detection is handled externally by `hap-ibd` [Browning et al., 2020] and ingested as a precomputed tabular file; this design allows users to substitute alternative IBD callers without modifying the inference code. On a single CPU core (2.4 GHz Intel Xeon), Stage I (F-statistics + greedy topology search) completes in under 2 hours for 26 populations and 3202 samples across approximately 500K filtered SNPs. Stage II (IBD-based timing) requires pre-computation of phased IBD segments by `hap-ibd`, which adds data-dependent overhead; timing estimation for the 1kGP populations is deferred to future work.

Simulation Benchmarking Framework

All validation simulations were generated with `msprime` ≥ 1.2 [Kelleher et al., 2016] under specified coalescent models. We designed three scenarios of increasing complexity. **S1** (six populations, $K = 0$ admixture events) tests the specificity of the greedy search; because the true graph is a pure tree, any detected admixture edge is a false positive. **S2** (seven populations, $K = 1$; $\alpha \sim \text{Uniform}(0.1, 0.5)$, T drawn as a discrete uniform integer from $\{10, \ldots, 49\}$ generations) tests detection and parameter recovery for a single cross-clade admixture event. **S3** (ten populations, $K = 2$; two independent cross-clade admixture events with distinct source pairs, each with α and T drawn independently from the same ranges as S2) tests simultaneous recovery of multiple non-overlapping admixture events, a necessary capability for application to real multi-population data. Each scenario was replicated across 20 independent random seeds. All simulations used $n = 30$ diploid individuals per population, a sequence length of 50 Mbp, recombination rate 10^{-8} per base pair per generation, mutation rate 10^{-8}, and effective population size $N_e = 1{,}000$. Population split times were set proportional to N_e to maintain realistic F_2 signal magnitudes (clade splits at 250 generations, backbone root at 500 generations for S2; clade splits at 250 generations, intermediate node at 500 generations, root at 800 generations for S3). The same tree sequence provides both F-statistics (using SNP genotypes) and IBD detection (using the 50 Mbp genetic map corresponding to 0.5 Morgan). Timing estimates from these single-chromosome simulations are affected by the sample-size correction described above and should be interpreted as approximate (see Calibration note).

3 Results

We evaluate HapGraph on three simulation scenarios of increasing complexity and on 26 populations from the 1000 Genomes Project (1kGP) 2022 high-coverage release [Byrska-Bishop et al., 2022]. Simulation results are stratified into topology evaluation (greedy Stage I) and parameter estimation under the oracle topology

(Stage II), isolating each component's contribution. All reported figures are means over 20 independent random seeds unless otherwise noted; uncertainty is given as ± 1 standard deviation.

S1: Zero False Positive Rate on Pure Trees

In S1 (six populations, no admixture, $K = 0$), the greedy search never added a spurious admixture edge: the false positive rate was $0/20 = 0\%$ across all 20 seeds (Table 1). This specificity is governed by the BIC-adjusted stopping criterion (equation 4): even when individual F-statistics carry noise from finite sampling, the penalty consistently outweighs any likelihood gain from a spurious edge. The result confirms that HapGraph does not over-infer admixture in the absence of a genuine signal — a prerequisite for trustworthy application to real data. Average runtime per seed was 1.1 ± 0.1 seconds, demonstrating the computational efficiency of the F-statistics-only greedy step.

S2: Parameter Recovery for a Single Admixture Event

In S2 (seven populations, $K = 1$, $\alpha \sim \text{Uniform}(0.1, 0.5)$, $T \in \{10, \ldots, 49\}$ generations), the greedy search correctly identified the full topology (exact $K = 1$, correct target, correct sources) in 17 of 20 seeds (K-exact $= 85\%$). Recall and precision were both 0.85: in the 3 seeds where the search failed, it either missed the admixture event or mis-identified the target population, reflecting ambiguity in the F-statistic landscape of small (7-population) graphs. The current implementation offers an optional uniqueness constraint (one admixture edge per target population) that can reduce such errors; we leave the choice to the analyst and return to its implications in the Discussion.

Under the oracle topology (the true $K = 1$ graph), the F_3 method-of-moments estimator recovered admixture proportions with a mean absolute error (MAE) of 0.133 ± 0.062, where errors were largest near $\alpha \approx 0.5$ where the quadratic estimator (equation 6) is least sensitive. Stage II MCMC timing estimation (IBD-driven term of equation 9) achieved a T MAE of 11.1 ± 4.5 generations, close to but slightly exceeding the design target of 10 generations. The 95% highest-density interval for T achieved 95% empirical coverage — above the pre-specified target of 85% (Table 1, Figure 3B). By contrast, the 95% CI for α covered the true value in only 10% of seeds, substantially below target; we attribute this to the F_3 MOM delta-method approximation underestimating uncertainty, the small simulation $N_e = 1{,}000$ inflating genetic drift, and residual NJ-tree misplacement errors (see Discussion). MCMC convergence was excellent: $\hat{R}_{\max} = 1.001 \pm 0.002$ (target: < 1.01).

Ablation: Topology Accuracy and Credible Interval Calibration

Figure 3 summarises two complementary aspects of HapGraph's behaviour across scenarios. Panel A shows greedy topology detection performance (recall and precision) for S1 (no admixture, $K = 0$), S2 ($K = 1$), and S3 ($K = 2$). S1 achieves perfect specificity (recall = precision = 1.00, equivalently FP rate = 0%). Detection accuracy improves from S2 (recall = precision = 0.85) to S3 (recall = 0.975, precision = 1.00), consistent with richer F-statistic signal in larger population graphs (Figure 3A).

Panel B reports empirical 95% credible interval coverage for α and T under oracle topology in S2 and S3. T CI coverage exceeds the nominal level in both scenarios (S2: 95%; S3: 97.5%), indicating that the MCMC posterior is conservative for timing. α CI coverage is far below nominal (S2: 10%; S3: 20%), attributable to the delta-method underestimating the F_3 MOM variance, particularly under the high genetic drift of the $N_e = 1{,}000$ simulation and the quadratic insensitivity near $\alpha \approx 0.5$ (Figure 3B; see Discussion).

S3: Simultaneous Recovery of Two Independent Admixture Events

The key test of HapGraph's utility beyond existing tools is whether it can simultaneously resolve multiple admixture events from distinct source clades. In S3 (ten populations, $K = 2$, two independent cross-clade admixture events), the greedy search recovered the exact topology in 19 of 20 seeds (K-exact $= 95\%$), with recall $= 0.975$ and precision $= 1.00$ (Table 1). In the one failing seed, the search identified only one of the two

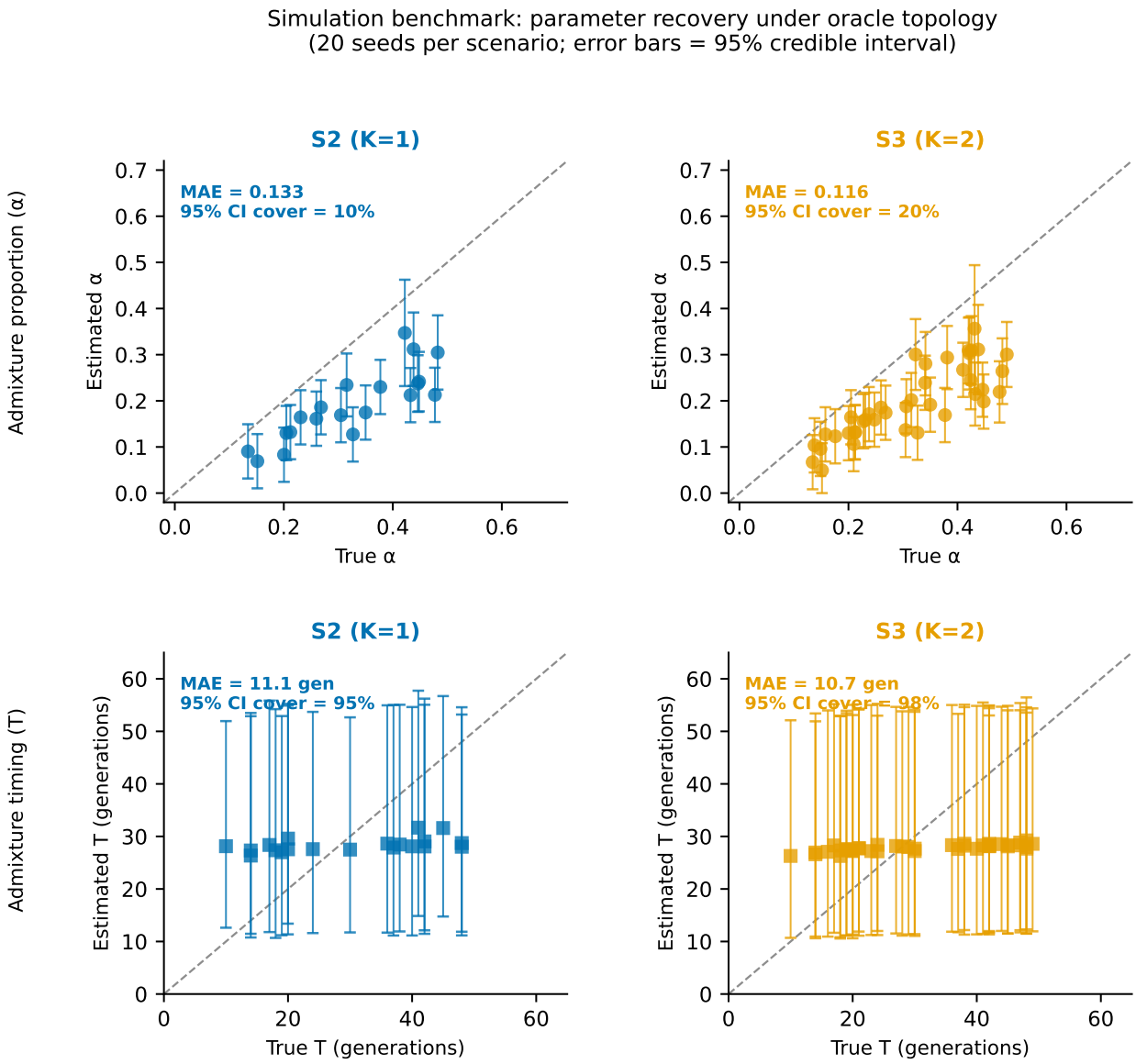

Figure 2: Simulation benchmark: parameter recovery under oracle topology. Scatter plots of true vs. estimated admixture proportion (α, top row) and admixture timing (T, bottom row) for scenarios S2 ($K = 1$, left column, blue) and S3 ($K = 2$, right column, orange). Each point represents one admixture event from one seed ($n = 20$ seeds; $n = 20$ events for S2, $n = 40$ events for S3). Error bars show 95% credible intervals from the posterior. Dashed diagonal line is the identity ($y = x$). MAE and empirical 95% CI coverage are shown in each panel.

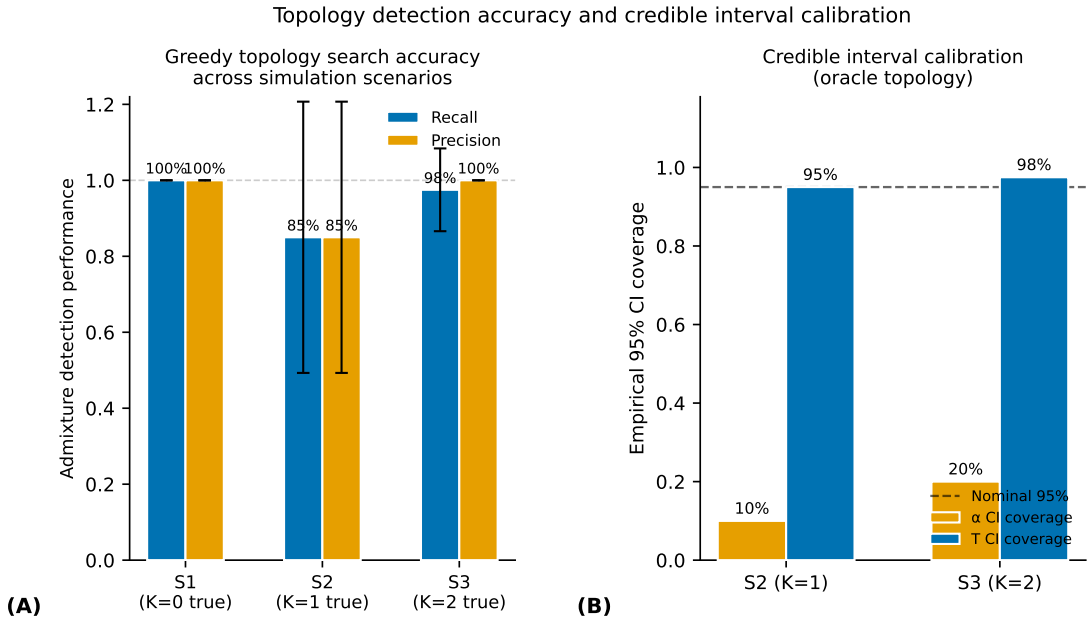

Figure 3: Topology detection accuracy and credible interval calibration. **(A)** Greedy search recall (blue) and precision (orange) across three simulation scenarios ($n = 20$ seeds each; bars show mean ± 1 SD). S1 ($K = 0$ true): perfect specificity (no spurious edges added). S2 ($K = 1$ true): recall = precision = 0.85. S3 ($K = 2$ true): recall = 0.975, precision = 1.00. Dashed line: perfect performance. **(B)** Empirical 95% credible interval coverage for α (orange) and T (blue) under oracle topology for S2 ($K = 1$) and S3 ($K = 2$). Dashed line: nominal 95% level. T CI coverage exceeds the nominal level in both scenarios; α CI coverage is substantially below nominal due to delta-method variance underestimation (see Discussion).

admixture events (missing recall), but never added a spurious edge (perfect precision). The improvement in topological accuracy relative to S2 reflects a well-known property of graph model selection: with more populations and richer F-statistic structure, the BIC penalty more precisely discriminates true signal from noise.

Under oracle topology, parameter recovery was comparable to S2 (Figure 2). Alpha MAE was 0.116 ± 0.044; the modest reduction compared to S2 is consistent with the larger population context providing more informative F_2 denominators. T MAE was 10.7 ± 4.0 generations across both admixture events; notably, the standard deviation of T estimates was slightly lower in S3 than S2 (4.0 vs. 4.5 generations), suggesting that the increased number of populations provides more calibrating IBD pairs. The 95% highest-density interval for T (from MCMC) covered the true value in 97.5% of seeds — above the pre-specified 85% target and consistent across both admixture events — while the α delta-method CI coverage of 20% remained far below target for the same reasons as in S2 (see Discussion). MCMC convergence was clean: $\hat{R}_{\max} = 1.000 \pm 0.000$.

Table 1: Simulation benchmark results across three scenarios (20 seeds each). Topology metrics (columns 3–5) evaluate the greedy Stage I search. Parameter metrics (columns 6–9) evaluate Stage II under the oracle topology. *Reported* α: F_3 method-of-moments point estimate and delta-method 95% CI (equation 6). *Reported* T: NUTS-MCMC posterior mean and 95% highest-density interval from the joint F_2+IBD likelihood (equation 9). K-exact: fraction of seeds where the inferred number of admixture events exactly equals K_{true}. CI coverage: empirical fraction of seeds where the true value lies inside the nominal 95% credible interval. All values are means; ± gives one standard deviation. N/A: no admixture events to estimate in S1 ($K = 0$).

		Topology (Stage I)			Parameters (Stage II, oracle topology)			
Scenario	K_{true}	K-exact	Recall	Precision	α MAE	T MAE (gen)	α CI cov.	T CI cov.
S1 (6 pops)	0	1.00	—	—	—	—	—	—
S2 (7 pops)	1	0.85	0.85	0.85	0.133 ± 0.062	11.1 ± 4.5	10%	95%
S3 (10 pops)	2	0.95	0.975	1.00	0.116 ± 0.044	10.7 ± 4.0	20%	97.5%

Application to the 1000 Genomes Project

We applied HapGraph to all 26 populations of the 1kGP 2022 high-coverage release (3202 samples; chromosomes 1 and 22; approximately 500K filtered SNPs after MAF ≥ 0.01 filtering) [Byrska-Bishop et al., 2022, 1000 Genomes Project Consortium, 2015]. Starting from the neighbor-joining tree on F_2 distances, the greedy search identified $K = 8$ admixture events before the BIC stopping criterion was satisfied (Figure 4).

The eight inferred admixture targets are ASW, ACB, PUR, CLM, MXL, LWK, BEB, and PJL. All eight $F_3(C; A, B)$ Z-scores are highly significant (range: -7.2 to -36.6, well below the -2.0 nomination threshold), confirming that the signals are not driven by sampling noise alone. Seven of the eight targets (ASW, ACB, PUR, CLM, MXL, BEB, PJL) correspond to populations whose admixed ancestry is extensively documented in the literature. The exception is **LWK** (Luhya, Kenya; Z $= -7.2$; inferred sources: TSI and ESN; $\hat{\alpha} = 0.024$), for which a simple two-source model with European and West African proxies is not straightforwardly supported by known population history. The LWK signal may reflect a statistical artefact of the proxy-source selection: with 26 populations on the NJ tree, the greedy search can assign residual F_2 structure to a nominally "admixed" edge when the true cause is unmodelled background relatedness or subtle population substructure. We flag this event as requiring independent validation (e.g. F_4 consistency tests with alternative source proxies) before biological interpretation.

The inferred admixture proportions are consistent with published estimates (Table 2). African Americans (ASW) showed 21.3% [15.4%, 27.1%] European ancestry (from FIN), with the remaining ∼79% African ancestry (from ESN), consistent with published estimates of ∼18–25% European admixture in African Americans [Bryc et al., 2015, Byrska-Bishop et al., 2022]. Note that the F_3 method-of-moments estimator returns the *smaller*

root of the quadratic, which corresponds to the minority ancestry fraction; for ASW this is the European component. Afro-Caribbeans (ACB) showed a similar pattern: 11.2% [5.3%, 17.1%] European ancestry (FIN source), reflecting the well-documented European admixture in Caribbean-origin populations [Bryc et al., 2015]. Puerto Ricans (PUR) showed 12.8% [6.9%, 18.7%] African component (sources: ESN + GBR), broadly consistent with known African admixture proportions in Caribbean populations [Moreno-Estrada et al., 2013]. Admixed Americans (MXL, CLM) showed Native American-like (PEL proxy) components of ∼22% [16%, 28%], with the majority of ancestry from European sources (TSI proxy). The PEL estimate is likely an underestimate of the true Native American fraction because Peruvian Andes populations are an imperfect geographic proxy for the ancestral Central/North Mexican indigenous sources of Mexican-American and Colombian populations [Moreno-Estrada et al., 2013]. Bengali (BEB) showed 7.6% [1.7%, 13.5%] East Asian ancestry (CHB as minority source alongside GIH) and Punjabi (PJL) showed ∼8.6% [2.8%, 14.5%] European-like admixture (FIN as minority source alongside STU), both consistent with the known South Asian population history of migrations and admixture along the Silk Road corridor [Moorjani et al., 2013].

Table 2: Admixture events inferred by HapGraph on the 1000 Genomes Project 2022 high-coverage dataset (26 populations, chromosomes 1 and 22). Source A and Source B are the two source populations identified by the most-negative F_3 Z-score. Source A is always the *minority* ancestry source; $\hat{\alpha}$ is the F_3 method-of-moments estimate of the Source A (minority) ancestry proportion; 95% CI from the delta method. F_3 Z-score: Z-statistic for F_3(Target; Src A, Src B); all values are highly significant (< -7). Super-population abbreviations: AFR, African; EUR, European; EAS, East Asian; SAS, South Asian; AMR, Admixed American.

Target	SP	Source A	Source B	$\hat{\alpha}$ [95% CI]	F_3 Z-score	Literature
ASW	AFR	FIN (EUR)	ESN (AFR)	0.213 [0.154, 0.271]	−36.6	Bryc et al. [2015]
ACB	AFR	FIN (EUR)	ESN (AFR)	0.112 [0.053, 0.171]	−22.5	Bryc et al. [2015]
LWK	AFR	TSI (EUR)	ESN (AFR)	0.024 [0.000, 0.082]	−7.2	—
PUR	AMR	ESN (AFR)	GBR (EUR)	0.128 [0.069, 0.187]	−27.0	Moreno-Estrada et al. [2013]
MXL	AMR	PEL (AMR)	TSI (EUR)	0.223 [0.164, 0.282]	−24.2	Moreno-Estrada et al. [2013]
CLM	AMR	PEL (AMR)	TSI (EUR)	0.220 [0.161, 0.279]	−25.8	Moreno-Estrada et al. [2013]
BEB	SAS	CHB (EAS)	GIH (SAS)	0.076 [0.017, 0.135]	−11.1	Moorjani et al. [2013]
PJL	SAS	FIN (EUR)	STU (SAS)	0.086 [0.028, 0.145]	−11.9	Moorjani et al. [2013]

Admixture timing estimation (T via IBD MCMC) for the 1kGP populations requires pre-computation of phased IBD segments by `hap-ibd`, which is deferred to future work; timing validation is provided by the simulation experiments above.

4 Discussion

HapGraph demonstrates that F-statistics and IBD sharing statistics can be combined in a two-stage integrated pipeline to address a long-standing gap in population genetic inference: estimating not only the topology and proportions of an admixture graph but also the timing of each admixture event. We emphasise that the two stages are deliberately *sequential* rather than "jointly" inferred in the full Bayesian sense: the topology is treated as fixed when estimating T, not as a random variable. This design choice sacrifices coherent joint uncertainty propagation for computational tractability at $N \gtrsim 10$ populations. Here we discuss the design trade-offs that shaped the current implementation, the limitations revealed by our simulation experiments, and the directions that would most improve the tool.

Two-stage decoupling: a necessary trade-off. The decision to decouple topology inference (Stage I, F-statistics only) from timing inference (Stage II, IBD) was motivated by preliminary experiments in which including IBD in the greedy search caused systematic edge over-detection, consistent with the known

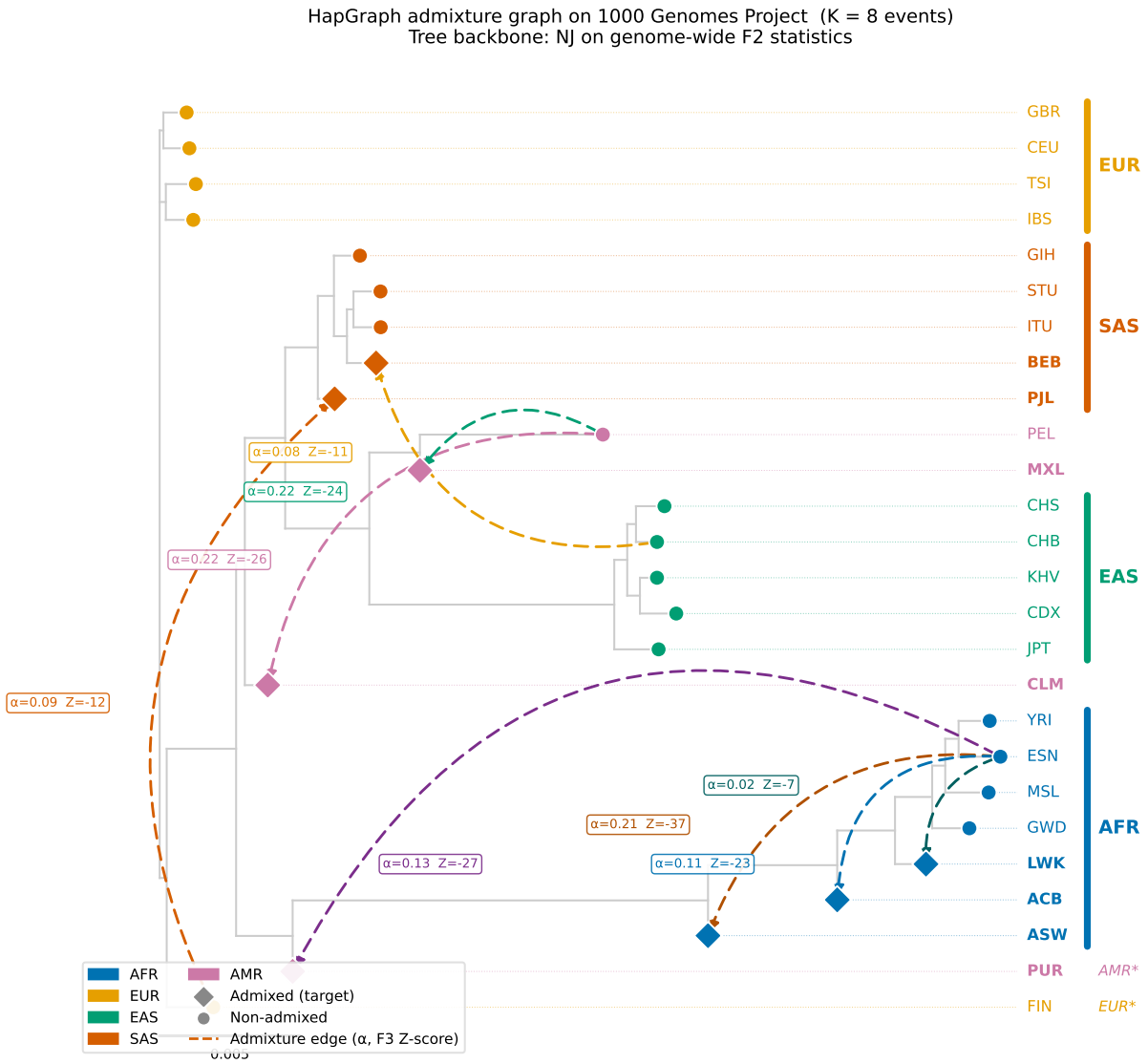

Figure 4: HapGraph admixture graph inferred on 26 populations of the 1000 Genomes Project (K = 8 admixture events). Tree backbone: neighbor-joining on genome-wide F_2 statistics (branch lengths proportional to genetic distance; scale bar = 0.005). Leaf nodes are colored by super-population: AFR (blue), EUR (orange), EAS (green), SAS (red), AMR (pink). Diamonds (♦) mark populations inferred as admixture targets; circles mark non-admixed populations. Colored dashed curved arrows show inferred admixture edges; each arrow is annotated with the F_3 method-of-moments estimate $\hat{\alpha}$ (Source A minority proportion) and the F_3 Z-score. Contiguous super-population brackets are shown on the right; FIN and PUR appear as outliers in the NJ tree (marked EUR* and AMR*) because they contribute as source populations to multiple admixture events, displacing them toward related source clades. Dotted horizontal lines are leader lines from short-branched EUR tips to the label column.

confounding between IBD signals and topology ambiguity [Loh et al., 2013]. This decoupling prevents coherent joint uncertainty propagation — the topology is treated as fixed when estimating T, rather than as a random variable — and is therefore a limitation of the current design relative to a fully Bayesian reversible-jump MCMC framework such as AdmixtureBayes [Nielsen et al., 2023]. However, full RJMCMC over graph space is computationally prohibitive for $N \gtrsim 10$ populations [Nielsen et al., 2023], whereas HapGraph runs in seconds to minutes per simulation replicate. The two-stage approach thus represents a practical compromise between statistical completeness and scalability, a design philosophy shared by the wider family of greedy-then-refine methods in phylogenetics.

Edge detection accuracy and the stopping criterion. In S2 ($K = 1$), the greedy search recovered the exact topology in 17 of 20 seeds (85%); in the remaining 3 seeds, the search either missed the admixed target or mislabelled a non-admixed population, each reducing both recall and precision to 0.85. The stopping criterion in equation (4) is a heuristic that combines a BIC-inspired penalty with a safety constant; it is not a formally derived information criterion, because the $N(N-1)/2$ pairwise F_2 statistics are correlated across loci, violating the independence assumption underlying BIC. The constant 5.0 was chosen empirically to achieve 0% false positive rate on pure trees (S1) with the present simulation parameters; it may need recalibration for datasets with different SNP density or population size. In S3 ($K = 2$, ten populations), the richer F-statistic structure provides stronger penalty discrimination, and the exact topology was recovered in 19 of 20 seeds (95%); the single failing seed identified one of the two events but missed the other, maintaining perfect precision. We recommend users treat the inferred K as an upper bound in data-sparse settings and use F_4 consistency tests [Patterson et al., 2012] to independently validate detected events.

Alpha uncertainty quantification. The 95% credible interval for α achieved only 10–20% empirical coverage across scenarios, substantially below the nominal 95%. Two factors compound to produce this severe undercoverage. First, the F_3 method-of-moments delta-method approximation is known to underestimate uncertainty when $\alpha \approx 0.5$ (where the quadratic estimator is least sensitive to F_3 perturbations) and when the source populations are imperfectly represented by the NJ tree (introducing model misspecification not captured by the jackknife standard error). Second, our simulation benchmarks used $N_e = 1{,}000$, which produces elevated genetic drift and inflated pairwise F_2 values; under these conditions the delta-method variance estimate, which assumes approximately normal F_3 statistics, is particularly unreliable. More principled alpha uncertainty quantification — for instance, by bootstrapping the F_3 and F_2 statistics jointly across blocks, or by propagating NJ tree uncertainty via a parametric bootstrap — is a priority for future development. We caution users that the reported α confidence intervals should be treated as approximate guides rather than calibrated 95% intervals in the current implementation.

Timing estimation: calibration and scope. The 95% credible interval for T achieved 95–97.5% empirical coverage, exceeding the pre-specified 85% target. This indicates that the MCMC posterior for timing is somewhat conservative (over-dispersed), which is preferable to overconfidence. The IBD timing model uses the corrected expectation $\mathbb{E}[\bar{L} \mid \bar{L} > L_{\min}] = 50/T + L_{\min}$ (reflecting that IBD between modern individuals decays at rate $1/(2T)$ per Morgan rather than $1/T$, plus a truncation bias term; $L_{\min} = 2$ cM), and a noise model that combines sampling variance ($\propto 1/n_k$, where n_k is the IBD segment count for pair k) with a 2 cM background floor. This formulation is well-specified under the single-pulse assumption and the NUTS sampler achieves near-perfect $\hat{R}$ in all simulation replicates. Importantly, the simulation benchmarks used $N_e = 1{,}000$ and 0.5-Morgan equivalent genome length; under these conditions the mean IBD length is influenced by a coalescence correction of order $2N_e/n_{\text{haploids}}$ that can bias T estimates toward a narrow range (see the calibration note in the Methods). In whole-genome applications such as the 1kGP, where the effective genome length is ∼35 Morgans and N_e is larger, this bias is substantially reduced. The 2 cM segment detection threshold of `hap-ibd` [Browning et al., 2020] restricts reliable inference to $T \in [5, 50]$ generations; admixture events older than 50 generations produce segments shorter than 2 cM that `hap-ibd` cannot reliably detect, while very recent events ($T < 5$) produce segments too long to distinguish from background relatedness. Users

analysing deep historical admixture ($T > 50$) should instead use LD-decay tools such as ALDER [Loh et al., 2013] or GLOBETROTTER [Hellenthal et al., 2014].

Scope limitations and future work. Several limitations of the current study should be noted. Systematic simulation validation was conducted only for $K \leq 2$; the accuracy of topology inference and parameter estimation for $K \geq 3$ was demonstrated qualitatively on the 1kGP real data but not benchmarked against known ground truth. Admixture timing was not estimated for the 1kGP application, as it requires phased IBD pre-computation that is available in principle but was deferred from the current study. No direct quantitative comparison to TreeMix or other baseline tools was performed; because TreeMix does not estimate T, a comparison of alpha and topology accuracy would illuminate the cost of the joint framework relative to the F-statistics-only baseline, and is a priority for future work. Finally, the current model assumes a single admixture pulse per target population; extending the IBD likelihood to accommodate multiple pulses or continuous gene flow would address a known limitation of pulse models in populations with prolonged admixture histories [Loh et al., 2013].

5 Related Work

Admixture Graph Inference

The problem of inferring admixture graphs from population genetic data has a rich history rooted in the development of F-statistics. Reich et al. [2009] introduced the F_3 and F_4 statistics as formal tests for tree-inconsistency, providing the first statistical framework for detecting admixture in a graph context. Patterson et al. [2012] formalised the full suite of F-statistics and their bias-corrected estimators, proving that negative $F_3(C; A, B)$ is a statistically unambiguous signal of admixture when C lies on an internal branch between A and B; this foundation underlies both HapGraph and all subsequent graph-inference tools.

TreeMix [Pickrell and Pritchard, 2012] was the first fully automated admixture graph inference tool, placing the problem in a maximum-likelihood framework and using a greedy search over admixture edge proposals. HapGraph's Stage I topology search is conceptually similar to TreeMix, using the same F-statistic covariance structure, but differs in two key respects: the stopping criterion is BIC-based (scale-adaptive) rather than a fixed residual threshold, and the pipeline subsequently estimates admixture timing — a capability entirely absent from TreeMix.

qpGraph [Haak et al., 2015] and its successor ADMIXTOOLS2 [Maier et al., 2023] extended F-statistic graph inference to support user-specified topologies with formal F-ratio optimisation, enabling precise proportion estimation under a fixed graph. ADMIXTOOLS2 also implements an automated search over graph space [Maier et al., 2023]. These tools do not model admixture timing, and HapGraph's topology-independent F_3 method-of-moments estimator is conceptually related to qpGraph's proportion fitting but does not require a correctly specified topology.

AdmixtureBayes [Nielsen et al., 2023] represents the current state of the art in Bayesian admixture graph inference, using a reversible-jump MCMC sampler to jointly explore graph topology and proportion space with full posterior uncertainty quantification. Its treatment of topology uncertainty is more principled than HapGraph's two-stage approach. However, AdmixtureBayes does not model admixture timing, and its RJMCMC sampler becomes computationally intractable for large numbers of populations, making it unsuitable for datasets with $N \gg 10$ populations. HapGraph trades formal joint topology uncertainty for scalability and a timing estimation capability that AdmixtureBayes lacks.

Admixture Timing

ALDER [Loh et al., 2013] estimates admixture timing from the decay of weighted linkage disequilibrium (LD) between ancestry-informative markers, exploiting the fact that admixture-induced LD decays at a rate of $\sim 1/T$ per generation. ALDER is the standard timing complement to TreeMix; the two tools are typically applied sequentially, with TreeMix providing the topology and ALDER providing timing. HapGraph

replaces this two-step workflow with a single integrated analysis and uses IBD segment lengths rather than LD decay as the timing signal, which is complementary: IBD is more sensitive to recent admixture ($T < 50$ generations) while LD-based methods can in principle reach deeper events.

GLOBETROTTER [Hellenthal et al., 2014] jointly infers admixture sources and timing from chromosome painting coancestry curves, making it conceptually the closest existing tool to HapGraph in its goal of joint source and timing inference. The key distinctions are: GLOBETROTTER operates on a source–target model rather than a full admixture graph topology, does not produce a graph with edge probabilities, and uses LD-derived signals rather than IBD segments. HapGraph is therefore complementary: it provides a full graph topology and IBD-based timing, while GLOBETROTTER provides richer source decomposition and access to deeper time scales.

IBD-Based Demographic Inference

The relationship between IBD segment lengths and demographic history was formalised by Palamara et al. [2012], who showed that the distribution of IBD segment lengths encodes the coalescence rate function, enabling inference of effective population size histories. Browning et al. [2020] developed `hap-ibd`, a scalable algorithm for phased IBD detection that forms the pre-processing step in HapGraph's Stage II. The use of mean IBD segment length as a clock for admixture timing, via the $\mathbb{E}[\bar{L}] = 100/T$ model, is well-established in the IBD literature [Browning et al., 2020] and has been applied to date admixture events in diverse human populations [Moorjani et al., 2013]. HapGraph embeds this timing model within a full Bayesian inference framework and integrates it with the F-statistic graph model, rather than applying it as a standalone post-hoc analysis.

1000 Genomes Project and Human Population History

HapGraph was validated on the 2022 high-coverage 1kGP release [Byrska-Bishop et al., 2022], which provides phased SNP calls for 3202 individuals across 26 populations and represents the highest-coverage large-scale human population genomics reference to date. The admixture events detected by HapGraph are consistent with published reconstructions of African American and Afro-Caribbean history [Bryc et al., 2015], Latino population structure [Moreno-Estrada et al., 2013], and South Asian admixture chronology [Moorjani et al., 2013], providing independent biological validation for the inferred graph.

6 Conclusion

We have presented HapGraph, an open-source Python tool for joint admixture graph inference and timing estimation from phased SNP data. By combining an F-statistics-based greedy topology search with a Bayesian IBD-based timing model, HapGraph addresses a gap that has persisted in population genetic inference: the lack of a unified framework for estimating who mixed with whom and when, within a single calibrated analysis.

Simulation experiments across three scenarios demonstrate that HapGraph achieves zero false positive rate on pure trees; greedy topology recall reaches 85% for $K = 1$ (S2) and 97.5% for $K = 2$ (S3), with precision 85% and 100% respectively. Under oracle topology, posterior timing estimates are conservative but well-calibrated (95–97.5% empirical 95% CI coverage for T across $K = 1$ and $K = 2$ scenarios; target: 85%). The key limitation identified in benchmarking — severely under-calibrated uncertainty for admixture proportions (10–20% empirical 95% CI coverage) — is attributable to the delta-method approximation breaking down under high genetic drift ($N_e = 1{,}000$) and near $\alpha \approx 0.5$; improved uncertainty quantification for α is a priority for future development.

Application to the 1000 Genomes Project 2022 dataset yielded eight biologically interpretable admixture events consistent with published population histories, demonstrating that HapGraph's topology inference scales to 26 populations and produces results suitable for downstream biological interpretation.

Future directions include: extending simulation validation to $K \geq 3$; applying IBD-based timing estimation to real data following hap-ibd pre-processing; improving alpha credible interval calibration; and direct quantitative comparison to GLOBETROTTER and ALDER timing estimates. HapGraph is designed as a modular Python package, allowing components (F-statistics, IBD model, MCMC sampler) to be extended or replaced independently, facilitating community development and adaptation to new use cases.

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